CareConnect — Patient-Provider EHR

Web-based Electronic Health Record System

Project Overview

CareConnect is a web-based healthcare platform that connects patients and doctors through a secure electronic health record system. It supports patient records, clinical documentation, prescriptions, appointments, CPOE/orders, and separate patient and doctor portals.

1. Main Modules

Module	Main Functions
User Authentication	Registration, login, logout, JWT authentication and role-based access.
Patient Management	Patient profile, personal details, basic health information and medical history.
Doctor Management	Doctor profile, specialization, assigned patients and patient search.
Medical Records	Diagnosis, consultation notes, medical history and clinical documentation.
Medication Management	Create and view prescriptions, medicine, dosage, frequency and duration.
CPOE / Orders	Doctors can create and manage laboratory tests and other clinical orders.
Appointment Management	Book appointments, view appointments and update appointment status.
Patient Portal	Patients can access their profile, records, prescriptions, appointments and reports.
Doctor Portal	Doctors can manage patients, records, prescriptions, notes and orders.
Admin Module	Manage patients, doctors and appointments with admin-only access.

2. Technology Stack

Layer	Technology	Purpose
Frontend	React.js + Vite	Build patient and doctor interfaces.
UI	HTML, CSS, JavaScript	Structure, styling and client-side functionality.
Routing	React Router	Navigation and protected role-based routes.
Backend	Node.js + Express.js	REST APIs and server-side application logic.
Database	MongoDB + Mongoose	Store users, patients, doctors, records, prescriptions and appointments.
Authentication	JWT + bcrypt	Secure login, password hashing and role-based access.
API Communication	REST API / Fetch or Axios	Communication between React frontend and backend.
Version Control	Git + GitHub	Source-code management and collaboration.

3. User Roles and Frontend

Patient Portal: Dashboard, profile, medical records, prescriptions, appointments, doctors and reports.

Doctor Portal: Dashboard, profile, patients, appointments, medical records, prescriptions, clinical notes and CPOE/orders.

Admin: Administrative access to manage users, doctors, patients and appointments.

4. System Test Cases

ID	Test Case	Sample Input	Expected System Response
TC-AUTH-01	Patient Registration	Name: Rahul Sharma; Role: Patient	Account is created and user can proceed to login.
TC-AUTH-02	Doctor Login	Valid doctor email and password	Credentials are verified and Doctor Dashboard opens.
TC-AUTH-03	Invalid Login	Wrong password	Login is rejected with an invalid credentials message.
TC-AUTH-04	Role-Based Access	Patient opens doctor-only route	Access is denied / 403 Unauthorized response.
TC-PAT-01	View Patient Profile	Patient selects Profile	Personal and basic health information is displayed.
TC-PAT-02	Update Profile	Phone and address	Profile is updated and new details are displayed.
TC-PAT-03	View Medical History	Patient selects Medical History	Available medical history is displayed.
TC-DOC-01	View Doctor Profile	Doctor opens Profile	Doctor name, specialization and contact details are displayed.
TC-DOC-02	View Assigned Patients	Doctor selects Patients	Assigned patient list is displayed.
TC-DOC-03	Unauthorized Patient Access	Patient calls doctor-only API	Backend rejects request because role is PATIENT.
TC-MED-01	Add Diagnosis	Patient P101; Viral Fever	Diagnosis is saved and linked to the patient record.
TC-MED-02	Add Consultation Note	Patient P101; Fever for 2 days	Clinical note is saved for authorized users.
TC-MED-03	Create Prescription	Paracetamol; 500mg; 2/day; 5 days	Prescription is saved and linked to the patient.
TC-MED-04	View Previous Records	Doctor opens Patient P101	Previous authorized records are displayed.
TC-APT-01	Book Appointment	Doctor D101; 30-09-2026; 10:30 AM	Appointment is created with the configured status.
TC-APT-02	View Appointments	Patient opens Appointments	Upcoming and relevant past appointments are displayed.
TC-APT-03	Update Appointment	PENDING → CONFIRMED	Appointment status is updated and reflected in UI.
TC-APT-04	Unavailable Slot	Already occupied time slot	Booking is rejected and slot-unavailable message is shown.
TC-ADM-01	Manage Patients	Admin opens Patient Management	Patient records and permitted actions are displayed.
TC-ADM-02	Manage Doctors	Admin searches D101	Doctor details and permitted actions are displayed.
TC-ADM-03	Manage Appointments	Admin searches A501	Appointment details and current status are displayed.
TC-ADM-04	Non-Admin Access	Doctor opens admin endpoint	Backend denies access because role is not ADMIN.

5. Example End-to-End Flow

Step	User Action	Expected Response
1	Patient registers with valid details.	Account is created successfully.
2	Patient logs in.	Patient Dashboard opens.
3	Patient books an available appointment.	Appointment is created.
4	Doctor logs in.	Doctor Dashboard opens.

Step	User Action	Expected Response
5	Doctor opens the assigned patient.	Authorized patient details and medical history are displayed.
6	Doctor adds diagnosis and clinical note.	Medical record is saved.
7	Doctor creates prescription.	Prescription is saved and linked to the patient.
8	Patient opens Medical Records.	Authorized medical information is displayed.

Note: The technology stack in this updated document follows the MERN-based implementation discussed for CareConnect: React, Node.js, Express.js and MongoDB.